

**MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL**

Paper Code : PCC-AI501/PCCAIDS501/PCCAIML 501 Probability and Statistics

UPID : 005934

Time Allotted : 3 Hours

Full Marks : 70

*The Figures in the margin indicate full marks.**Candidate are required to give their answers in their own words as far as practicable***Group-A (Very Short Answer Type Question)**

1. Answer any ten of the following :

[1 x 10 = 10]

- (i) The partial differential equation corresponding to $z = f\left(\frac{xy}{z}\right)$ is
- (ii) If X and Y are two independent random variables then $E(XY) = ..$
- (iii) Name a family of parametric distribution in which mean = variance .
- (iv) Using Lagrange's method solve the following partial differential equation: $yzp + zxq = xy$.
- (v) The distribution for which mean is always greater than variance is.....
- (vi) Which distribution possessing the memoryless property ?
- (vii) Find the general integral of the linear partial differential equation $p-q=\log(x+y)$.
- (viii) Binomial distribution with parameters n and p is said to be symmetric if.....
- (ix) What is a significance level?
- (x) Formulate the PDE from $z=f(x+iy)+g(x-iy)$
- (xi) Three companies A, B and C supply 25%, 35% and 40% of the notebooks to a school. Past experience shows that 5%, 4% and 2% of the notebooks produced by these companies are defective. If a notebook was found to be defective, what is the probability that the notebook was supplied by A?
- (xii) If t is a statistic having standard error 3 then find the variance of t ?

Group-B (Short Answer Type Question)

Answer any three of the following :

[5 x 3 = 15]

2. Find the Mean and variance of Poisson Distribution. [5]
3. Eliminate the arbitrary functions f and F from $y=f(x-at)+F(x+at)$. [5]
4. If the independent random variables X, Y are binomially distributed, respectively with $n=3$, $P=1/3$, and $n=5$, $P=1/3$, write down the probability that $X+Y \geq 1$. [5]
5. In a sample of 1000 people, 540 were rice eaters and the rest were wheat eaters. Can we assume that the proportion of rice eaters is more than 50% at 1% level of significance. [5]
6. If $Z=aX+bY$ and r is the correlation coefficient between the random variables X and Y, find variance of Z. [5]

Group-C (Long Answer Type Question)

Answer any three of the following :

[15 x 3 = 45]

7. (a) If a continuous random variable X has exponential distribution with parameter λ then find the mean and variance of the distribution. [8]
- (b) If a random variable has the gamma distribution with parameter $l=2$ and $\lambda=1/3$ find the mean and the standard deviation of this distribution. Hence find the probability that the random variable will take a value less than 5. [7]
8. (a) The standard deviation of symmetrical distribution is 3. What must be the value of the fourth moment about the mean in order that the distribution be mesokurtic? [7]
- (b) If the first four moments of distribution about the value 5 are equal to -4, 22, -117 and 560 determine the corresponding moments about the mean [8]
9. (a) Solve $\frac{y-z}{yz}p + \frac{z-x}{zx}q = \frac{x-y}{xy}$. [7]
- (b) Solve $p+3q=5z+\tan(y-3x)$. [8]

10. (a) The distribution of a random variable X is given by $P(X=-1)=1/8$, $P(X=0)=3/4$, $P(X=1)=1/8$. Verify Chebyshev's inequality for the distribution. [5]
- (b) Show by Chebyshev's inequality that in 2000 throws with a coin the probability that the number of heads lies between 900 and 1100 is at least $19/20$. [5]
- (c) Suppose the p.d.f of a random variable X is [5]

$$f(x) = \frac{(x+1)}{2}, -1 < x < 1$$

and $f(x)=0$ otherwise. Find $E(X)$ and $\text{Var}(X)$. [8]

11. (a) Solve $y \frac{\partial u}{\partial x} + x \frac{\partial u}{\partial y} + 2xy = 0$; $u(s, 2s) = 2s^2$ [7]
- (b) Solve the partial differential equation $p - q = 2yu$

*** END OF PAPER ***